A Spectral–Spatial Feature Rotation-Based Ensemble Method for Imbalanced Hyperspectral Image Classification

Yi Su, Xiaojun Li, Junping Yao, Chengrong Dong, and Yao Wang, *Member, IEEE*

***Abstract*— Hyperspectral image classification has two key characteristics: spatial dependency and class imbalance. The synthetic oversampling methods for solving the between-class imbalance have the problem of spatial information loss to some extent. However, the feature space optimization methods for solving data within-class complex distribution problems have not been studied extensively. To remedy such an issue, we propose in this article an improved spectral–spatial feature rotation-based support vector machine (SVM) ensemble method (SSFRoS). More precisely, in the data preprocessing stage and from the perspective of sample space optimization, we devise a spectral–spatial neighborhood-based adaptive synthetic sampling algorithm (SSAS) that focuses adaptively on hard-to-learn samples, with artificial samples interpolated in the spectral–spatial neighborhood to further utilize the spatial information. In the feature rotation stage and from the perspective of feature space optimization, we innovatively discover that the class overlapping problem can be handled by combining feature extraction, to achieve the goal of reducing the difficulty of imbalanced classification. Besides, the SVM and voting procedure are used in the ensembled classification stage. We then conduct a series of experiments to compare the proposed SSFRoS with several baselines in terms of seven metrics. The results show that SSFRoS has significant advantages in dealing with imbalanced hyperspectral image classification problems. In addition, a detailed ablation study shows that the lowest class recall can be improved by 46.67% and 5.26% at Indian Pines and Pavia University, respectively, which demonstrates that using SSAS can enhance the recognition of hard-to-learn classes.**

***Index Terms*— Class overlapping, feature rotation, hyperspectral image, imbalanced classification, resampling.**

I. Introduction

HYPERSPECTRAL imaging, a technology combining imaging and spectral detection techniques, uses different sensors to detect the electromagnetic waves reflected from objects [1]. It covers continuous spectral information from hundreds to thousands of narrow bands while collecting image information of each pixel in the target area [2], [3]. Classification, one of the main tasks in hyperspectral image processing, aims to assign a unique class label to each observed target (pixel). With the increased demand for military reconnaissance, precision agriculture, environmental monitoring, resource exploration, medical diagnosis, and other applications, the classification of hyperspectral images has become a research hotspot in image processing and pattern recognition [4], [5], [6], [7]. Previous hyperspectral classification studies are usually based on the assumption of pixel independence. Individual pixels are split into independent spectral domain vectors, each element being a measurement corresponding to a specific wavelength. Due to the inherent properties of hyperspectral data, there are two key characteristics in its classification: spatial dependency and class imbalance.

For spatial dependency, since the original hyperspectral image is a 3-D data cube, there are spatial structural correlations between the spectral features and between the class of each pixel in the image, that is, pixel dependency and label dependency between adjacent pixels [8]. The independent vector representation method leads to the loss of spatial structure information [9], and the classification that considers only the spectral features of the vectors is known as pixel-wise classification. However, the spatial position of each pixel obviously carries important information about its classification. For example, adjacent pixels are more likely to be different parts of the same object. Pixels of the same material will have similar spectral properties, so more likely to be regarded as the same class, but with spatial information such as different object sizes, they can be separated into different classes. The design of the feature representation determines how much information it contains about the observed data. Therefore, the discriminability of feature representation directly determines the upper bound of classification performance [10]. Many studies in recent years have shown that the spatial information introduced through hyperspectral spatial dependency in classification can effectively improve the performance compared to the traditional pixel-wise classification, which is called spectral–spatial classification [8], [11], [12], [13].

Class imbalance means that the samples (pixels) of hyperspectral images have the inherent characteristics of imbalance since the number, size, and spatial distribution of objects are different. The explicit representation is that most samples belong to some classes while a few samples belong to other classes. The basic problem of imbalanced classification is that

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due to the large variation in the number of data samples of different classes, most standard learning algorithms tend to pay more attention to the majority samples while ignoring the classification accuracy of the minority samples, to optimize the overall classification error [14]. As a result, some algorithms have poor performance in recognizing the minority samples, though they can achieve high overall accuracy on imbalanced datasets. In practice, however, we usually place more emphasis on the accurate identification of the minority samples. For example, for the detection of a particular target, the pixels occupied by the target are much less than those occupied by the background of land, trees, and so on. It is more important to correctly identify the pixels of the target than to correctly identify the pixels of other objects.

Existing research on imbalanced hyperspectral classification generally adopts a combination of resampling strategies to solve the imbalance between the number of samples of different classes. In the process of synthetic oversampling, the search and selection of neighbors are based on K nearest neighbors (KNN), without considering the spatially adjacent samples (spatial nearest neighbors), which to some extent results in a lack of spatial information on synthetic samples. However, the feature space optimization methods of imbalanced data have received less attention. Among them, the improved ensemble method based on the rotation forest (RoF) uses the feature extraction algorithm to generate diversified feature spaces. But it considers only the contribution of the feature variance but not the within-class complex distribution of imbalanced data to optimize the feature space. Although it is effective to improve the overall classification performance, it is ineffective in the imbalanced classification problem. In view of the above problems, this article proposes a spectral–spatial feature rotation-based support vector machine (SVM) ensemble method (SSFRoS) which is applicable to imbalanced hyperspectral classification. It is a three-stage method consisting of data preprocessing (feature fusion and resampling), feature rotation, and ensembled classification. Our experiments prove that it can reduce the classification performance gap between classes and effectively solve the imbalanced classification problem while maintaining superior overall classification performance. The main contributions of this article are as follows.

1) A spectral–spatial neighborhood-based adaptive synthetic sampling algorithm (SSAS) is proposed. It is an imbalanced hyperspectral classification preprocessing algorithm considering both spectral and spatial features. Based on the fusion of features and from the perspective of sample space optimization, it focuses adaptively on the sampling of hard-to-learn samples. The improved algorithm determines hard-to-learn samples based on the spectral–spatial neighbor. It considers not only the sample distribution in the KNN neighborhood, but also the proportion of heterogeneous training samples in the spatial neighborhood. Furthermore, it carries out interpolation synthesis of samples in the spectral–spatial neighborhood, which further introduces spatial information for the synthetic samples.
2) On the basis of RoF, an improved rotating ensemble method based on feature optimization for class overlapping is proposed. From the perspective of feature space optimization, it is innovatively proposed to find a rotating feature space that can alleviate class overlapping in the within-class complex distribution, to reduce the difficulty of imbalanced classification. Specifically, combined with the feature extraction step, subspace projections with a larger between-class distance and a smaller within-class distance are found for the randomly split feature subsets. A modified rotation matrix calculation method is used to solve a variety of optimized feature spaces and further improve the overall classification performance and alleviate imbalanced classification problems.

The remainder of this article is organized as follows: Section II provides an overview of related work. Section III describes the proposed method in detail. The data, configuration, and results of the experiments are discussed in Section IV. Finally, the work of this article is summarized in Section V.

II. Related Work

A. Research on the Essence of Imbalanced Classification

The imbalance between the number of samples of different classes is often referred to as a between-class imbalance. However, many studies have found that between-class imbalance is not necessarily associated with the degradation of classification performance in practice, which indicates that there are other hidden factors that make it more difficult to classify imbalanced data [15], [16]. These hidden factors include data complexity problems that cannot be observed, that is, the distribution of real data often has such problems as small disjuncts, noise, and class overlapping [17], [18], [19]. Disjunct means that data are not continuously distributed in the data space, but is intermittently distributed in subclusters. Subclusters containing only a small number of samples are called small disjuncts. Disjuncts are widespread, but due to the relatively small number of minority class samples in the imbalanced distribution, the samples of subclusters gathered by separation are more sparse, and small disjuncts are more likely to occur. Classifying data with disjuncts can be understood in a way that the classifier will try to describe the class with multiple disjuncts by creating rules with multiple disjuncts [19]. Standard classifiers usually classify large disjuncts accurately, but it is difficult to use extremely limited information to build rules for classifying the subclusters when learning small disjuncts with too sparse samples. The noise is not necessarily related to the imbalance of data, but the boundary between excessively sparse subclusters and noise is blurred. The clusters of these samples are subclusters with entity existence, or the noise is difficult to identify. Some subclusters without enough samples may be mistaken for noise and misclassified [20]. Class overlapping problem refers to that samples of different classes have high similarity in some features, resulting in overlapping distribution areas in the feature space. Prati et al. [16] conducted classification experiments on the synthetic artificial dataset, and the results proved that: when the distance between-classes is far, the classification difficulty is low, and even if the imbalance between-classes is high, the classification performance will

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not be significantly affected. When the problem of class overlapping and class imbalance is intertwined, the difficulty of classification is significantly increased. Among them, the influence of class overlapping is even stronger than that of between-class imbalance.

The solutions to the imbalanced classification problem can be divided into two main categories: 1) data-level methods, which refer to the modification of the original dataset through data preprocessing, including the optimization of the sample space and the feature space; and 2) algorithm-level methods, which refer to the improvement or ensemble of classification algorithms. In addition, it can also mix the two level methods.

B. Data-Level Methods

In data-level methods, the resampling methods of sample space optimization focus on the explicit problem of between-class imbalance, which can solve or alleviate the poor classification of a few classes in most cases, and they can be further classified into under-sampling, oversampling, and hybrid sampling. Synthetic oversampling rebalances the dataset by generating synthetic minority samples, which can increase the number of minority samples and add the necessary information for learning, thus improving the classification accuracy of minority classes and the generalization ability of the classifier. Currently, the synthetic minority oversampling technique (SMOTE) [21] and its variations are widely used. In essence, synthesizing new minority samples through the SMOTE algorithm is performing random interpolation in the nearest neighbor samples based on the KNN algorithm. Several studies have optimized the seed sample selection strategy. Borderline-SMOTE [22] selected minority samples that are closer to the decision boundary between two classes to generate synthetic samples. Adaptive synthetic sampling approach (ADASYN) [23] generated weights based on the sample distribution, to use the minority class samples which are surrounded by majority classes in feature space to generate more synthetic samples.

In the current hyperspectral imbalance classification research, the resampling method is widely used for preprocessing. Özdemir et al. [24] used different resampling algorithms to rebalance the hyperspectral data fed into the convolutional neural network (CNN) model. Li et al. [25] proposed an orthogonal complementary subspace projection (OCSP)-based bidirectional resampling algorithm, choosing representative samples of majority classes, and synthesizing new minority samples in a nonlinear way. Subudhi et al. [26] optimized the percentage of oversampling required for each class by using the particle swarm optimization algorithm (PSO). However, due to the direct combination of SVM, extreme learning machine (ELM), and KNN classification algorithms, some results were slightly inferior to those of other more complex classification algorithms. When feature extraction and classification in deep-learning methods are implemented end-to-end, making it difficult to integrate the oversampling steps after feature extraction [27]. A common solution is to combine the encoder–decoder framework with classification, using the deep-learning model to generate minority samples that are as close to the real data distribution as possible [28]. Zhu et al. [29] proposed a spectral–spatial-dependent global learning (SSDGL) framework, the hierarchically balanced (H-B) sampling strategy is used to resample the training data by class, and the weighted softmax loss is optimized by gradient descent method to reweight each class sampling rates, to pay more attention to the samples that are incorrectly classified. Xi et al. [30] proposed a deep generative spectral–spatial classifier (DGSSC), including a two-stage encoder, decoder, and classifier. According to the potential variables obtained by the encoder, the decoder can achieve minority-class data augmentation and can use the end-to-end method training to optimize the joint loss of the three components. Roy et al. [31] proposed a tensor-based 3-D-HyperGAMO model, using generator and discriminator networks to oversample 3-D hyperspectral patches of the minority class, with both the original and generated samples used to train the 3-D classifier. However, when the number of minority classes is too small, synthetic oversampling is likely to generate highly repetitive samples, which may not be applicable to extremely imbalanced data. When the within-class complexity such as small disjuncts, noise, and class overlapping is high, more noisy samples may be generated and class overlapping may be increased, which can hardly achieve good results. In contrast, the feature space optimization method focuses more on solving the implicit problem of data complexity. From the perspective of finding a feature space more conducive to classification, it solves the problem that the minority class cannot be well represented by the original space and weakens the imbalanced classification problem on most datasets. Although classic feature extraction algorithms such as principal component analysis (PCA), minimum noise separation, Fisher discriminant analysis (FDA), and independent component analysis have been widely used in the design of hyperspectral feature representation, the ability of some algorithms to alleviate the problem of imbalanced data complexity has not been widely concerned.

C. Algorithm-Level Methods

Among algorithm-level methods, the common ones include one-class learning, cost-sensitive learning, threshold-moving method, and ensemble learning [20]. One-class learning is a singular point detection method that uses only one class of samples to complete the learning. When it is used in imbalanced classification, there are problems such as wasting precious minority samples, high quality requirements about training set such as no outliers and noise, and only suitable to binary classification. Cost-sensitive learning aims to correct the equal cost of implicit assumptions in standard classification algorithms by setting the misclassification cost of specific classes, but the cost matrix in actual classification problems is often unknown [32]. Threshold-moving refers to changing the probability that a sample is predicted to be a specific class so that the sample is more cautiously discriminated from the minority class and achieves the purpose of correcting the bias of the standard algorithm toward the majority class. However, it is difficult to solve the closed-form solution of the thresholds for optimization targets such as the F_1 score in multiclassification [33]. In practice, the optimization thresholds are usually

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set empirically, since search-based methods may lead to overfitting problems. Ensemble learning combines different base classifiers to achieve better classification performance, robustness, and generalization than individual classifiers [34], [35]. Therefore, it can alleviate the poor performance of a single weak classifier on minority classes and achieve better results in imbalanced classification than using traditional classification algorithms directly [33]. According to the method of base classifiers combination, it can be divided into three categories: parallel, cascade, and hybrid. The parallel ensemble framework has no interaction between parallel base classifiers and relatively low computational complexity. The accuracy and diversity of the base classifiers are usually considered two key factors for ensemble learning. The classical bagging (bootstrap aggregating) algorithm [36] introduces diversity by sampling the dataset with the bootstrap algorithm. Based on this, the RoF algorithm [37], [38] introduces more diversity by using the PCA feature extraction algorithm to rotate the feature subspace through random feature subset splitting. Many researches have achieved further improvements in the RoF for hyperspectral classification applications [39], [40]. However, most studies were conducted without considering the effects of data imbalance factors. For the imbalanced classification problem, the combination of data-level method and algorithm-level ensembled classifier is widely used. SMOTE-bagging [41] is a method that incorporates the resampling strategy into the ensemble framework, and its main idea is to use the dynamically varying resampling rate for first sampling before each base classifier and then synthesize minority samples using the SMOTE algorithm.

In recent years, some researchers have studied the ensemble methods of hyperspectral imbalance classification. Lv et al. [42] proposed a depth ensemble algorithm, enhanced-random-feature-subspace-based ensemble CNN (EECNN), which randomly oversampled and multiply enhancing training samples based on random feature subspace, and then constructed an ensembled classifier by using CNN as the base classifier. Miftahushudur et al. [43] modified the distance function used in Safe-Level SMOTE to measure nearest neighbors with the Manhattan distance, and a differentiable sampling rate (DSR) is adopted to ensemble deep-learning classifiers. Quan et al. [44] proposed a novel double-ensemble algorithm, which first evaluates the feature importance by an ensemble model, and then used oversampling to ensemble base classifiers. Feng et al. [45] proposed DS-RoF that introduced dynamic synthetic minority oversampling to the RoF. AdaSRoF proposed by Feng et al. [46] is based on the combination of variable sampling rate resampling and RoF. It uses the serial ensemble strategy to select the samples at the current decision boundary with a higher probability in the SMOTE process of the next base classifier to generate synthetic samples, solving the problem of reducing the noise of synthesized samples. The above two methods performed synthetic oversampling and subsequent classification based on the spectral feature representation without introducing spatial information. Quan et al. [47] proposed a weighted depth rotating forest SMOTE-WDRoF, which combined the spatial neighborhood information extraction strategy with SMOTE method. It fed the balanced data into the RoF and the subsequent cascade forest and dynamically increased the weights of misclassified samples according to the classification results of each level. However, the multilayer algorithm of cascade design was much more complex.

The imbalanced hyperspectral classification methods based on ensemble learning mostly combine resampling strategies and rarely consider the optimization of feature space. And the improved ensemble method based on RoF, although using the PCA feature extraction algorithm in standard RoF, generates diversified feature spaces from the perspective of variance contribution of features. It does not consider solving the problem of within-class data complexity of imbalanced data itself or achieving the feature space optimization of imbalanced distribution.

In general, it has become a consensus to use spectral–spatial classification methods that consider interpixel structural information in hyperspectral classification. However, the existing imbalanced hyperspectral classification research generally combines the resampling strategy, and there is still the problem of spatial information loss in the process of synthetic oversampling. Simple and effective, the resampling method is used extensively and is an acceptable solution to the imbalance problem. However, when the number of minority classes is too small and the complexity of the within-class is high, it may not achieve desirable results. Therefore, it is necessary to conduct in-depth research on feature space optimization and algorithm-level optimization. In the imbalanced hyperspectral classification method based on RoF, although feature extraction is carried out, the feature space is not optimized by combining the within-class imbalance distribution of imbalanced data.

III. Proposed Method

To address the problems of existing research, specifically make full use of spatial information in the hyperspectral synthetic minority oversampling step and reduce the difficulty of imbalanced classification by alleviating the problem of class overlapping, this article proposes an SSFRoS. SSFRoS is inspired by 2-D-singular spectrum analysis (SSA) [48], ADASYN [23], and DS-RoF [45], and it consists of three stages—data preprocessing (feature fusion and resampling), feature rotation, and ensembled classification—as shown in Fig. 1. In this section, vectors or matrices are written in bold letters and scalars are written in regular letters.

A. Data Preprocessing (Feature Fusion and Resampling)

Data preprocessing is a stage that performs feature fusion and resampling, in which the minority class in the original training set is synthetically oversampled according to the set sampling strategy in the fused spatial–spectral feature space to obtain a balanced sample set. Fig. 2 gives the feature fusion steps and sampling strategy of the SSAS algorithm. In pixel-level classification, the algorithm based on KNN is directly used for random interpolation of the sample's neighbors in the feature space (hereinafter referred to as the feature neighbors). It does not consider the sample's neighbors in the space

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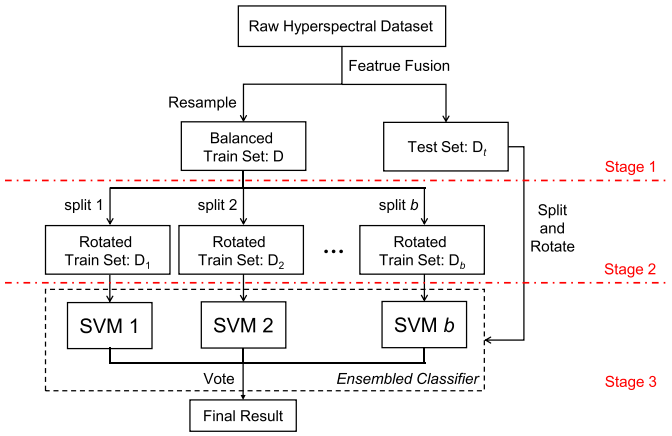


Fig. 1. Spectral–spatial feature rotation-based ensemble method.

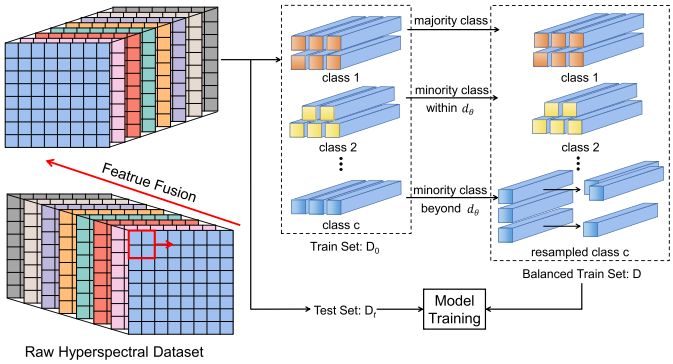


Fig. 2. Schematic of SSAS feature fusion and resampling algorithm.

(hereinafter referred to as the space neighbors) and abandons the spatial structure of hyperspectral images. The main idea of the proposed SSAS is to first fuse spectral features and spatial features of samples through a 2-D-SSA algorithm, and then an improved spectral–spatial neighborhood-based method is used to enhance the oversampling of hard-to-learn samples and generate synthetic samples with spatial neighbor information. When calculating the sampling rate, not only the labels of feature neighbors are considered, but also the labels of spatial neighbors in the training set are counted. Those surrounded by heterogeneous samples are more likely to be selected as seed samples, and the neighbor samples are randomly selected from the feature neighborhood and spatial neighborhood of the seed samples, then the synthetic samples are generated by interpolation. Using the method of considering two types of neighborhoods, it can not only make the synthetic samples carry the information of adjacent samples in the space, but also increase the diversity of the synthetic samples, and further introduces the spatial information in the balanced sample set.

1) Spectral–Spatial Feature Fusion: The essence of this step is to achieve feature extraction of the spatial information between the pixel samples by the 2-D-SSA algorithm and to integrate the spatial 2-D features into the spectral features of each pixel sample. Since there is no spatial structure relationship between the real sample and the synthetic sample, the spatial structure must be extracted first. SSA is a 1-D time-series analysis and prediction method which has been proven effective in many fields. For 2-D images on hyperspectral arbitrary bands, the original features can be grouped and reconstructed using 2-D-SSA [49], an extended version of SSA on 2-D. In addition to feature fusion, it can also handle noise by filtering components with a large contribution [48]. The 2-D image on a certain band can be expressed as an image matrix of $N_x \times N_y$ shape.

The embedding step involves scanning the 2-D image using a sliding window and embedding the obtained spatial structure information into the trajectory matrix. Define the window size as $L_x \times L_y$, where $L_x \in [1, N_x]$, $L_y \in [1, N_y]$. The window is used to scan the 2-D image $\mathbf{E}$ from left to right and from top to bottom, and e represents the spectral value. When the given reference point at the upper left corner is (i, j), the window information obtained by scanning can be expressed as the window matrix $\mathbf{L}_{i,j}$

$$\mathbf{L}_{i,j} = \begin{pmatrix} e_{i,j} & e_{i,j+1} & \cdots & e_{i,j+L_y-1} \\ e_{i+1,j} & e_{i+1,j+1} & \cdots & e_{i+1,j+L_y-1} \\ \vdots & \vdots & \ddots & \vdots \\ e_{i+L_x-1,j} & e_{i+L_x-1,j+1} & \cdots & e_{i+L_x-1,j+L_y-1} \end{pmatrix}. \quad (1)$$

The operator vec indicates the vectorization of the matrix, that is, the matrix is pieced together line by line, then the trajectory matrix $\mathbf{T}$ can be expressed as

$$\mathbf{T} = \begin{pmatrix} \text{vec } \mathbf{L}_{1,1} \\ \text{vec } \mathbf{L}_{1,2} \\ \vdots \\ \text{vec } \mathbf{L}_{1,N_y-L_y+1} \\ \vdots \\ \text{vec } \mathbf{L}_{N_x-L_x+1,1} \\ \text{vec } \mathbf{L}_{N_x-L_x+1,2} \\ \vdots \\ \text{vec } \mathbf{L}_{N_x-L_x+1,N_y-L_y+1} \end{pmatrix}^T. \quad (2)$$

The trajectory matrix of the 2-D signal has a Hankel–block–Hankel structure (HBH), which means that the trajectory matrix $\mathbf{T}$ can be represented as a Hankel matrix of the sub-matrix. Each submatrix $\mathbf{H}_b$ conforms to the Hankel structure

$$\mathbf{T} = \begin{pmatrix} \mathbf{H}_1 & \mathbf{H}_2 & \cdots & \mathbf{H}_{N_x-L_x+1} \\ \mathbf{H}_2 & \mathbf{H}_3 & \cdots & \mathbf{H}_{N_x-L_x+2} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{H}_{L_x} & \mathbf{H}_{L_x+1} & \cdots & \mathbf{H}_{N_x} \end{pmatrix}_{L_x \times (N_x-L_x+1)} \quad (3)$$

$$\mathbf{H}_b = \begin{pmatrix} e_{b,1} & e_{b,2} & \cdots & e_{b,N_y-L_y+1} \\ e_{b,2} & e_{b,3} & \cdots & e_{b,N_y-L_y+2} \\ \vdots & \vdots & \ddots & \vdots \\ e_{b,L_y} & e_{b,L_y+1} & \cdots & e_{b,N_y} \end{pmatrix}_{L_y \times (N_y-L_y+1)}. \quad (4)$$

In the singular value decomposition (SVD) step, eigen decomposition is first conducted on matrix $\mathbf{T}\mathbf{T}^{\mathrm{T}}$, and the calculated eigenvalues are represented as $\lambda_1, \lambda_2, \ldots, \lambda_r$ from the largest to the smallest. The corresponding eigenvectors are $\mathbf{u}_1, \mathbf{u}_2, \cdots, \mathbf{u}_r$, respectively, where r indicates the rank of $\mathbf{T}$. Then the singular value expansion can be obtained, and the items in the formula are defined as the component matrix

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(or elementary matrix) $\mathbf{T}_k$

$$\mathbf{T} = \sum_{k=1}^{r} \sqrt{\lambda}_k \mathbf{u}_k \mathbf{v}_k^{\mathrm{T}} \stackrel{\mathrm{def}}{=} \sum_{k=1}^{r} \mathbf{T}_k \tag{5}$$

$$\mathbf{v}_k = \frac{\mathbf{T}^{\mathrm{T}} \mathbf{u}_k}{\sqrt{\lambda}_k}. \tag{6}$$

The contribution (the amount of information carried) of each component $\mathbf{T}_k$ to $\mathbf{T}$ can be measured by the corresponding eigenvalue λ_k.

The grouping step aims to divide the r components obtained from SVD decomposition into $\mathbf{M}$ disjoint subsets $\mathbf{G}_m$, $m \in [1, M]$. The grouping matrix $\mathbf{T}_{Gm}$ is defined as

$$\mathbf{T}_{Gm} \stackrel{\mathrm{def}}{=} \sum_{k \epsilon \mathbf{G}_m} \mathbf{T}_k = \sum_{k \epsilon \mathbf{G}_m} \sqrt{\lambda}_k \mathbf{u}_k \mathbf{v}_k^T. \tag{7}$$

The contribution of each grouping can be measured by the sum of the eigenvalues corresponding to the components within the group. In this article, to extract the main spatial structure information and effectively filter noise, the grouping principle is retaining larger contribution grouping.

The reconstruction step refers to the case where the selected grouping matrix combination does not necessarily still have the HBH structure so that the matrix can be reconstructed by two-step Hankelization to regain the characteristic. Thus, the 2-D signal can be reconstructed. First, each submatrix is restored to the Hankel structure by the diagonal average operation on the subdiagonal, and then the submatrix is Hankelized in the grouping matrix. We number the minor diagonals of the matrix from top left to bottom right and use $\tilde{a}_d$ to refer to the element on the dth subdiagonal line after Hankelization. Then, the process of Hankelizing $m \times n$ type ($m \leq n$) matrix $\mathbf{A}$ is defined as follows:

$$\tilde{a}_d = \begin{cases} \dfrac{1}{d} \displaystyle\sum_{i=1}^{d} b_{i,d-i+1}, & 1 \leq d < m \\ \dfrac{1}{m} \displaystyle\sum_{i=1}^{m} b_{i,d-i+1}, & m \leq d \leq n \\ \dfrac{1}{m+n-d} \displaystyle\sum_{i=d-n+1}^{m} b_{i,d-i+1}, & n < d \leq m+n-1. \end{cases} \tag{8}$$

In the second step of Hankelization, the average of the subdiagonal is still carried out according to the above equation, and the diagonal average between the submatrices is the average of the corresponding elements. The obtained HBH matrix is used to reconstruct the 2-D signal. First, each submatrix is vectorized, and then the obtained row vectors are spliced according to the subdiagonal order to obtain the reconstructed new 2-D image matrix of $N_x \times N_y$. The 2-D image reconstruction on all bands is completed, that is, the feature extraction of spatial structure is completed, and the spatial information is fused into the spectral feature to obtain the spectral–spatial feature.

2) Calculating the Sampling Probabilities Based on Spectral–Spatial Neighborhoods: Set the corresponding classes of pixel sample vectors $\mathbf{x}_i$ as $y_i \in \{1, 2, \cdots, c\}$, where c denotes the total number of classes. The adaptive sampling probability of SSAS determines whether a particular sample is a hard-to-learn sample by measuring how many heterogeneous samples are in its spectral–spatial neighborhood and thus adjusts the number of synthetic samples generated with it as a seed sample. The class with the largest number of samples is defined as the majority class l and all the rest as the minority class. N_j represents the number of samples in class j. Then, the imbalance degree d_j of class j can be defined as

$$d_j \stackrel{\mathrm{def}}{=} \frac{N_j}{N_l} \tag{9}$$

where $d_j \in (0, 1]$. A threshold $d_\theta \in (0, 1]$ is set for the tolerable degree of imbalance. When $d_j < d_\theta$, the imbalance degree of the class is high and oversampling is required. Then, the oversampling rate $\beta \in [d_\theta, 1]$ is set for controlling the number of samples g_j synthesized in the class

$$g_j = \left(N_l - N_j\right) \times \beta. \tag{10}$$

When $d_\theta = 1$, all classes whose number is not perfectly balanced between classes need to be oversampled. $\beta = 1$ means that a perfectly balanced dataset is generated after oversampling. The Euclidean distance of the vectors is used to calculate k_e featured neighbors of $\mathbf{x}_i$, and Δ_e indicates the number of samples within the featured neighborhood that are different from the class of $\mathbf{x}_i$. The window with the size of $w \times w$ is used to select the (w^2-1) nearest spatial neighbors of $\mathbf{x}_i$ on the 2-D image, and Δ_a indicates the number of training samples within the spatial neighborhood that are different. In the experiment of this article, when a hyperspectral image is used for tenfold cross-validation, the training data is not a complete image, and some samples not included in the training set are not taken into account. Only k_a spatial neighbors in the training set are used for calculation. Although the feature representation after feature fusion carries spatial information to a certain extent, the synthetic oversampling algorithm based on KNN only interpolates samples and their neighbors in the feature space, and it does not directly utilize the feature dependence between spatial neighbor samples. To further generate synthetic samples carrying spatial information, increase the diversity of synthetic samples generated by interpolation in the neighborhood, and alleviate the problem of excessive repetition of synthetic samples that may occur when the training samples are small, SSAS also takes the samples of the featured neighborhood and the spatial neighborhood into consideration. For the convenience of description, the above two types of neighborhood are referred to as the spectral–spatial neighborhood in this article. When oversampling a class, it is necessary to calculate the proportion of heterogeneous samples within the spectral–spatial neighborhood of the sample $\mathbf{x}_i$, and then the sampling probability $\hat{r}_i$ of the sample $\mathbf{x}_i$ being selected as the seed sample is obtained by normalizing all samples within the class

$$r_i = \frac{\Delta_e + \Delta_a}{k_e + k_a} \tag{11}$$

$$\hat{r}_i = \frac{r_i}{\sum_{i=1}^{N_j} r_i}. \tag{12}$$

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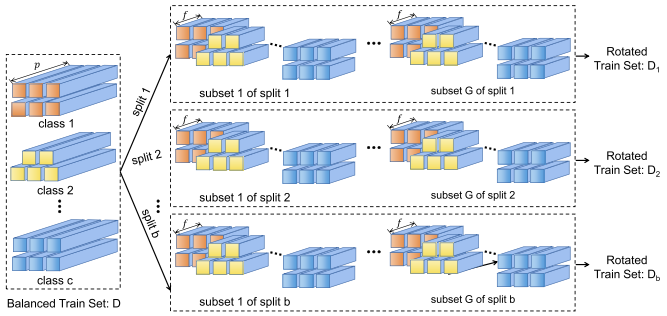


Fig. 3. Schematic of subset division and feature rotation.

3) Generating Synthetic Samples: The strategy for generating synthetic samples in this article follows from the standard SMOTE algorithm, where a homogeneous sample $\mathbf{x}_u$ is randomly selected within its spectral–spatial neighborhood after the seed sample $\mathbf{x}_i$ is selected, and random interpolation is performed using random values, ran $\in [0, 1]$, to obtain synthetic samples $\mathbf{x}_i^u$:

$$\mathbf{x}_i^u = \mathbf{x}_i + (\mathbf{x}_i - \mathbf{x}_u) \times \text{ran}. \tag{13}$$

The number of synthetic samples g_j^i that need to be generated using $\mathbf{x}_i$ as seed sample and belong to class j can be calculated

$$g_j^i = g_j \times \hat{r}_i. \tag{14}$$

B. Feature Rotation

The goal of this stage is to reduce class overlapping of imbalanced data and reduce the difficulty of imbalanced classification by the method of feature space optimization, to alleviate the low performance of minority classes. Meanwhile, it provides different data inputs for the base classifiers in the subsequent ensemble stage by finding diversified spaces. The feature rotation method modified in this article first splits the balanced sample set into feature subsets randomly, and then we innovatively discover that the class overlapping problem can be handled by combining feature extraction. The projection matrix of feature subsets is solved through the FDA algorithm, and the modified projection vector feature alignment method is employed to build a sparse rotation matrix $\mathbf{R}$. The original dataset can be transformed into new feature space through a dot-by-rotation matrix. As a result, diversity can be introduced to the ensembled classifier through multiple random subset splits. Fig. 3 illustrates the subset split and feature rotation.

The FDA algorithm aims to find the best projection vector, which can achieve minimum within-class distance and maximum between-class distance after the original sample vector is projected on the vector. Therefore, the problem of class overlapping can be alleviated and the classification task is relatively easy. Let f indicate the dimension of the feature subset and $\mathbf{x}_i$ the vector of the sample on this feature subspace. Then, the projection vector of the solution $\mathbf{w} = (a_1, a_2, \ldots, a_f)^T$ is the vector of the same dimension. According to the FDA criterion, $\mathbf{w}$ is the eigenvector of matrix $\boldsymbol{S}_b\boldsymbol{S}_{\boldsymbol{w}}^{-1}$. The larger the eigenvalue, the more favorable is the feature space obtained with the corresponding eigenvectors as projection vectors for classification. $\boldsymbol{S}_w$ denotes the within-class scatter matrix, which is obtained by summing the covariance matrix of various classes, that is, the sum of within-class distance measurement reflects the degree of hash among each within-class in the sample set. $\boldsymbol{S}_b$ indicates the between-class scatter matrix, which measures the distance between the centroids of the classes and the centroids of all samples to reflect the degree of hash among the between-classes. Since the rank of $\boldsymbol{S}_b$ is $c - 1$ at most, as many as $m = \min\{f, c - 1\}$ features can be obtained through feature extraction. According to reduced dimensions m, the corresponding eigenvectors are solved in the order of eigenvalues from large to small and obtain the eigenmatrix $\mathbf{W} = (\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_m)$.

During the feature rotation of the balanced dataset, the new features of each dimension in each sample are obtained by multiplying the sample vector with any projection vector, and the features of each dimension in the sample are multiplied by the corresponding elements of the projection vector when the vectors are multiplied. Essentially, it is a random feature selection and corresponding subspace transformation of each sample in turn. In this article, we propose a new feature matrix construction method while retaining the original randomness and considering the information utilization. Specifically, G feature matrixes ($f \times m$) are first obtained through random subset split, which makes the number of nonzero elements in the constructed matrix $\mathbf{R}$ less than or equal to that in the matrix constructed by the RoF algorithm. Then, on the basis of aligning the elements of the feature vector in the column, the elements of the feature vector are randomly rearranged in the row direction by using the randomness of the feature subset split, to perform random feature selection with respect to the reduced dimensions m. The specific steps of solving the rotation matrix $\mathbf{R}$ for each random split are as follows.

1) Initialize the rotation matrix $\mathbf{R}$ to an all-zero square matrix of $p \times p$.
2) 75% of the training samples are randomly sampled to get a new training set.
3) For each feature subset, solve for the within-class scatter matrix $\boldsymbol{S}_w$ and between-class scatter matrix $\boldsymbol{S}_b$.
4) Solve for all eigenvalues and corresponding eigenvectors of the square matrix $\boldsymbol{S}_b\boldsymbol{S}_{\boldsymbol{w}}^{-1}$ and obtain the eigenvectors $\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_m$ in the order of eigenvalues from largest to smallest. Then the eigenmatrix $\mathbf{W}$ composed of them is

$$\mathbf{W} = (\mathbf{w_1}, \mathbf{w_2}, \ldots, \mathbf{w_m}) = \begin{pmatrix} \boldsymbol{a_{11}} & \cdots & \boldsymbol{a_{1m}} \\ \vdots & \ddots & \vdots \\ \boldsymbol{a_{f1}} & \cdots & \boldsymbol{a_{fm}} \end{pmatrix}. \tag{15}$$

5) Perform feature alignment. The feature full set index corresponding to feature u in the feature subset is indicated by index_u, and then a_{uv} will be placed in the

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row index$_u$ and column index$_v$ of the R matrix

$$\mathbf{R} = \begin{array}{c} \\ \\ \\ \\ \text{index}_u \end{array} \overset{\text{index}_v}{\begin{pmatrix} \vdots & \vdots & \vdots & \\ \cdots\ a_{11} & \cdots\ a_{1v} & \cdots\ a_{1m} & \cdots \\ \vdots & \vdots & \vdots & \\ \cdots\ a_{u1} & \cdots\ a_{uv} & \cdots\ a_{um} & \cdots \\ \vdots & \vdots & \vdots & \\ \cdots\ a_{p1} & \cdots\ a_{pv} & \cdots\ a_{pm} & \cdots \\ \vdots & \vdots & \vdots & \end{pmatrix}}. \tag{16}$$

6) Repeat steps 2–4 for all feature subsets of this division to obtain the final rotation matrix $\mathbf{R}$.

The random sampling in step 2 is to avoid obtaining the same rotation matrix due to accidentally obtaining the same subset of splits in different splits, and it can further increase the randomness and reduce the training time. Set the rotation matrix obtained by the bth split as $\mathbf{R}_b$, and the data matrix of the balanced sample set $\mathbf{X} = (\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N)^T$ is transformed to obtain the rotation sample set $\mathbf{X}_b$

$$\mathbf{X}_b = \mathbf{X}\mathbf{R}_b. \tag{17}$$

C. SVM Ensembled Classifier

The working principle of a rotation-based ensemble classifier is to train different base classifiers by transforming the samples into different feature spaces and then fuse the results of each base classifier with the majority voting method

$$H(\mathbf{x}) = \underset{j}{\mathbf{argmax}} \sum_{b=1}^{B} h_b^j(\mathbf{x}) \tag{18}$$

where $h_b^j(\mathbf{x})$ denotes the output of the base classifier h_b on class j. In this article, SVM is chosen as the base classifier.

IV. Experiments and Result Analysis

A. Datasets

To verify the effectiveness of the SSFRoS, two well-known datasets with class imbalance problems are selected. The Indian Pines dataset has an image size of 145 × 145, and the remaining 10 249 pixels containing surface features are labeled as 16 classes. The imbalance ratio (ratio between the largest and smallest classes) is 122.75. There are 220 bands, and a corrected version consisting of 200 bands after removing the water absorption bands is generally used. To measure the proposed method comprehensively and examine its robustness to noise, experiments are conducted on both the corrected and uncorrected versions of the Indian Pines dataset in this article. The Pavia University dataset has an image size of 610 × 340, with 42 776 pixels containing surface features. They are labeled into nine classes with an imbalance rate of 19.69. There are 103 bands remaining after 12 bands were removed due to noise effects. The data description and ground truth are shown in Table I and Fig. 4.

B. Experimental Settings

The experiments in this article are implemented using MATLAB and python. To study the imbalance classification problem, the mean and standard deviation of the tenfold stratified cross-validation are used as the results. A series of experiments are conducted for the Indian Pines dataset with the train size as 30%, 20%, and 10%. For the Pavia University dataset, the train size is set as 10%, 5%, and 1%. For the baseline, the following methods are adopted: 1) two classical classification algorithms commonly used, SVM and RoF. 2) The 2-D-SSA feature extraction algorithm is used and combined with the KNN classification algorithm (performs better than the combination with SVM) be abbreviated as 2-D-SSA, while combined with the RoF classification algorithm be abbreviated as SSARoF. 3) DSRoF, which is designed for imbalance hyperspectral classification. 4) The 3-D-HyperGAMO is a newly reported hyperspectral imbalance classification deep learning method. Due to huge differences in model structure and complexity, we made a separate comparison and analysis between the baseline and the proposed method.

In the experiment, the ensemble method based on feature rotation needs to set the number of base classifiers B and the number of feature subsets G. Methods use PCA such as RoF and DSRoF take $B = 30$ and $G = 30$ according to the original setting. SSFRoS based on feature optimization for class overlapping proposed in this article takes $B = 10$. According to the different classification number required by the Indian Pines dataset and the Pavia University dataset, the number of split subsets for the two datasets is $G = 15$ or $G = 10$, respectively. The window size is set as 10 × 10 for the 2-D-SSA step on the sliding window. When calculating neighborhoods, the number of featured neighbors is set as $k_e = 7$ and the window size of the spatial neighborhood is set to 5 × 5. For the resampling of this article, the imbalance threshold and sampling rate are set to 0.8. For the DSRoF algorithm using dynamic sampling rate, the range of sampling parameters is set to be [10%, 20%, 30%,..., 100%] in line with the original paper. For the methods using the SVM classifier, the RBF kernel is selected and a one-for-one strategy is used to achieve multiple classifications. For 3-D-HyperGAMO, the size of the spatial window is set to 11 × 11 × bands, and according to the hardware we use, the batch and learning rate are set to 300 and 0.0005, respectively. Under this setting, after 300 iterations, better results can be obtained than the original report. It is necessary to be noted that since the complexity of 3-D-HyperGAMO is significantly higher than that of other methods, we have replaced computer equipment with a higher performance, which is composed of an NVIDIA Tesla T4 processor with 16 GB GPU memory and 128 GB RAM, the computer equipment used for other experiments is composed of an Intel i7-11800H processor with 16 GB RAM. The experimental results (metrics and running time) in this article do not exactly match the results given in the original paper. It may be due to the use of different programming languages and dependency libraries, the use of the tenfold stratified cross-validation, and the difference in the setting of some parameters not specified in the original paper. Algorithms should be

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numbered and include a short title. They are set off from the text with rules above and below the title and after the last line.

C. Evaluation Metrics

Seven metrics are selected for evaluation, and the model training time (TIME) is given as a reference for algorithm complexity. In addition, to measure the influence of different methods on the class overlapping problem, a distance metric Dis is proposed to quantify the severity of the class overlapping problem.

1) Overall Accuracy (OA): Overall accuracy is also commonly referred to as accuracy or OA. It can reflect the overall classification performance of the classifier, but not the performance of imbalance data classification.

2) Class-Recall, MinRecall, and Recall Range (R-Range): Class-recall, class-precision, or class-F_1 are usually used instead of class-accuracy due to the different number of samples in each class. Since Type I errors may cause more serious consequences than Type II errors in minority classification, recall tends to be more important. Therefore, this article uses class-recall to measure the classification performance of each class.

MinRecall is the lowest value of class-recall, which mainly reflects the classification performance on the most hard-to-learn class and reflects the lower limit of the classification method. The recall range (abbreviated as R-range below) is the extreme difference in class-recall, which reflects the difference between the classification performance in each class. It can effectively measure the effectiveness of methods in dealing with imbalance classification problems.

3) Average Metrics: Average metrics is the average of precision, recall, F_1 score, and so on. This evaluation metric is independent of the sample size of each class and is important for evaluating the performance of methods in imbalance classification. In contrast to class-accuracy, precision, recall, F_1 score, and other metrics are also used in some research to calculate average accuracy. It can be defined as

$$\text{Average Metric} = \frac{1}{C}\sum_{i=1}^{c}\text{Metric}_i. \tag{19}$$

4) Distance Metric Dis Reflecting the Severity of Overlapping Problem: The severity of class overlapping in the binary second classification can be quantified by the ratio of the distance between the centroids of the classes and the standard deviation of features between samples [16]. Accordingly, the ratio of the distance measurement between the centroids of the classes and the distance measurement between samples can be used as a quantitative metric for the problem of class overlapping in multiclass high-dimensional data of realistic distribution. To measure the impact of each method on the class overlapping problem, this article uses the ratio Dis of the average distance between the centroids of the classes D_c and the average distance from the sample to the sample centroid D_s as a quantitative metric. The effectiveness of the feature spaces solved by methods is evaluated as follows:

$$\boldsymbol{\mu} = \frac{1}{N}\sum_{i=1}^{N}\mathbf{x}_i \tag{20}$$

TABLE I
DATA DESCRIPTION OF THE INDIAN PINES AND PAVIA UNIVERSITY DATASETS

	Indian Pines		Pavia University	
No.	Class	Samples	Class	Samples
1	Alfalfa	46	Asphalt	6631
2	Corn-notill	1428	Meadows	18649
3	Corn-mintill	830	Gravel	2099
4	Corn	237	Trees	3064
5	Grass-pasture	483	Painted metal sheets	1345
6	Grass-trees	730	Bare Soil	5029
7	Grass-pasture-mowed	28	Bitumen	1330
8	Hay-windrowed	478	Self-Blocking Bricks	3682
9	Oats	20	Shadows	947
10	Soybean-notill	972		
11	Soybean-mintill	2455		
12	Soybean-clean	593		
13	Wheat	205		
14	Woods	1265		
15	Buildings-Grass-Trees-Drives	386		
16	Stone-Steel-Towers	93		
Total		10249		42776

$$D_c = \frac{1}{c(c-1)}\sum_{j=1,k\neq j}^{c}\left|\boldsymbol{\mu}_j - \boldsymbol{\mu}_k\right| \tag{21}$$

$$D_s = \frac{1}{N}\sum_{i=1}^{N}|\mathbf{x}_i - \mu| \tag{22}$$

$$\text{Dis} = \frac{D_c}{D_s} \tag{23}$$

where $\boldsymbol{\mu}$ represents the sample centroid vector. Similarly, the vector $\boldsymbol{\mu}_j$ representing the jth class centroid can be obtained.

D. Results and Analysis

1) Classification Performance Analysis: Tables II–IV show the experimental results of the tenfold cross-validation on the corrected version of the Indian Pines dataset (train size = 30%, 20%, and 10%) through baselines and SSFRoS proposed in this article, respectively. The tables list the mean value and standard deviation of the six metrics, including OA, class-recall, R-range, AP, AR, and AF_1. The MinRecall is marked in bold, and the best method for each metric is underlined. If the values of one metric are the same, the class or method with a small standard deviation is better. To visually show the differences between each method and highlight the areas of misclassified samples, the classification map of each method when train size = 10% is drawn in Fig. 5.

Overall, the SSFRoS method proposed outperforms baselines in most metrics, and the advantages in corresponding metrics can also be clearly observed from the classification maps. It has better class-recall value in most classes and its R-range value is significantly lower than those of all baselines. The largest values are obtained in OA and average metrics. It indicates that SSFRoS has a better performance in the classification of different classes, and the difference between classes is smaller. As a result, SSFRoS is a method with

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TABLE II
EVALUATION METRICS (%) BY DIFFERENT METHODS ON THE CORRECTED INDIAN PINES DATASET WITH TRAIN SIZE = 30%

Class No.	SVM	RoF [39]	2D-SSA [49]	SSARoF	DSRoF [46]	SSFRoS
1	**0.00±0.00**	66.88±11.14	92.19±7.25	**93.13±5.86**	65.94±8.65	97.50±2.47
2	54.92±2.48	81.05±1.57	93.44±1.35	98.07±0.86	72.01±1.80	99.71±0.23
3	43.75±2.14	71.07±2.42	96.44±1.22	98.67±0.95	67.92±1.96	99.86±0.14
4	14.16±4.12	75.90±5.23	99.10±1.03	99.88±0.25	76.33±3.30	99.82±0.57
5	81.57±2.87	91.92±1.12	95.62±1.25	98.85±0.81	92.22±1.64	99.97±0.09
6	97.79±0.52	97.75±0.44	98.26±0.71	98.00±0.69	97.16±1.04	99.78±0.25
7	**0.00±0.00**	84.50±4.97	92.00±10.33	97.50±4.25	85.50±8.64	100.00±0.00
8	99.46±0.34	99.34±0.59	99.79±0.25	99.85±0.25	99.46±0.42	100.00±0.00
9	**0.00±0.00**	**43.57±14.46**	**50.71±17.96**	93.57±9.19	**62.14±11.19**	100.00±0.00
10	62.04±3.76	84.12±1.54	91.63±1.27	98.18±0.99	79.03±1.58	99.85±0.20
11	89.91±1.36	91.05±0.79	96.57±0.70	99.38±0.35	92.76±0.87	97.61±0.53
12	37.42±5.06	79.66±2.19	88.75±2.40	98.51±1.25	80.58±1.90	99.16±0.68
13	95.80±2.08	97.20±1.78	99.44±0.44	99.72±0.36	97.20±2.06	99.72±0.68
14	98.00±0.31	97.02±0.82	98.56±0.69	99.76±0.20	95.32±1.34	99.95±0.06
15	32.41±3.17	66.41±3.56	97.26±1.83	99.19±0.49	70.11±2.82	99.85±0.36
16	87.23±2.41	94.62±2.83	94.92±3.98	94.46±4.12	96.62±3.06	**96.92±4.35**
R-range	99.46±5.06	55.77±14.02	49.08±17.71	6.75±8.99	37.32±10.77	3.08±4.35
OA	72.58±0.63	87.02±0.44	95.68±0.26	98.84±0.18	85.46±0.31	99.24±0.13
AP	65.00±0.56	89.32±0.90	92.85±1.69	98.58±0.53	88.99±0.58	99.18±0.20
AR	55.90±0.60	82.63±1.11	92.79±1.05	97.92±0.91	83.14±0.97	99.36±0.29
AF_1	57.21±0.57	85.01±1.01	92.57±1.15	98.21±0.62	85.35±0.74	99.26±0.18
TIME(s)	0.31±0.02	18.50±0.42	33.99±0.71	49.89±0.79	326.14±5.28	77.67±0.78

TABLE III
EVALUATION METRICS (%) BY DIFFERENT METHODS ON THE CORRECTED INDIAN PINES DATASET WITH TRAIN SIZE = 20%

Class No.	SVM	RoF	2D-SSA	SSARoF	DSRoF	SSFRoS
1	**0.00±0.00**	58.92±8.04	78.92±12.27	85.68±13.30	65.68±6.50	96.49±4.94
2	47.29±3.01	79.38±0.81	91.00±1.67	97.02±0.91	68.18±3.09	99.46±0.33
3	40.86±1.71	69.01±2.94	95.08±1.11	98.40±0.85	64.50±1.83	99.56±0.56
4	12.89±2.80	67.26±4.74	98.42±1.45	98.84±1.16	73.74±4.99	99.42±0.68
5	76.22±5.12	89.35±3.76	93.52±2.40	98.29±1.22	90.52±3.28	99.72±0.36
6	97.79±0.71	97.28±0.55	97.89±1.20	98.22±0.86	96.37±1.33	99.91±0.15
7	**0.00±0.00**	83.18±8.85	92.27±9.83	95.00±9.92	82.27±7.86	100.00±0.00
8	99.29±0.25	99.06±0.73	98.46±1.76	99.14±1.19	99.06±0.76	100.00±0.00
9	**0.00±0.00**	**35.00±7.91**	**32.50±11.71**	**83.13±11.04**	**61.88±9.97**	98.75±2.64
10	52.57±5.91	83.08±1.31	89.11±2.01	97.16±1.28	76.05±2.06	99.81±0.24
11	90.07±1.01	90.11±1.10	94.50±1.15	99.12±0.29	92.90±0.59	96.81±0.85
12	21.03±4.64	75.33±2.75	86.65±2.21	97.79±1.22	75.64±3.40	99.07±0.91
13	94.39±2.05	96.89±2.95	99.21±0.29	99.45±0.45	96.16±2.59	99.82±0.29
14	98.06±0.36	96.33±0.52	97.27±1.17	99.55±0.55	95.04±0.69	99.96±0.05
15	21.26±1.84	64.53±2.31	95.40±1.23	99.00±0.93	67.83±2.06	99.81±0.27
16	84.05±2.61	90.14±4.77	94.46±2.58	92.84±7.40	95.27±3.00	**95.41±5.49**
R-range	99.29±5.91	64.06±8.33	66.71±11.98	16.42±13.01	37.18±9.38	4.59±5.49
OA	68.70±1.05	85.42±0.49	93.84±0.48	98.29±0.18	83.73±0.39	98.94±0.24
AP	62.66±0.95	87.78±1.33	89.42±1.61	97.38±0.95	87.06±0.80	98.61±0.77
AR	52.24±0.89	79.68±1.05	89.67±1.11	96.16±1.39	81.32±1.09	99.00±0.28
AF_1	53.35±0.91	82.41±1.06	89.15±1.22	96.65±1.03	83.47±0.79	98.76±0.49
TIME(s)	0.12±0.01	11.34±0.23	33.99±0.71	44.55±0.89	153.97±3.67	56.97±0.56

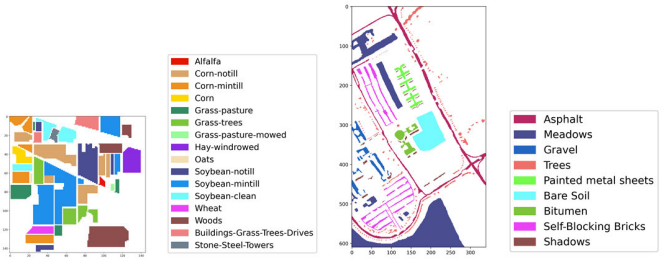
excellent overall performance for handling imbalance classification problems. Taking the RoF as the standard, SSFRoS does not degrade the overall performance when narrowing the gap between classes.

DS-RoF is designed for the imbalanced classification problem. Compared with the standard RoF, it improves MinRecall from 43.57% to 62.14% at train size = 30%, and its R-range decreases from 55.77% to 37.32%, indicating that it does play a certain effect on the imbalance problem. In comparison, the MinRecall value of SSFRoS is 96.92% and the R-range is 3.08%, which means that it further solves the classification performance imbalance between classes. As the train size decreases to 10%, the performance of all methods decreases. The MinRecall value of DS-RoF is 18.33% and the R-range is 80.20%. This confirms one of the motivations of this article: when the minority samples used for training are few, the

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TABLE IV
EVALUATION METRICS (%) BY DIFFERENT METHODS ON THE CORRECTED INDIAN PINES DATASET WITH TRAIN SIZE = 10%

Class No.	SVM	RoF	2D-SSA	SSARoF	DSRoF	SSFRoS
1	**0.00±0.00**	44.15±14.15	62.93±8.03	73.17±14.77	51.22±14.18	93.41±5.64
2	42.23±3.61	75.42±3.28	84.58±2.25	94.51±1.46	59.50±3.54	98.50±0.67
3	28.54±10.29	63.35±2.73	87.86±2.17	95.31±1.72	56.56±4.03	97.66±1.82
4	6.71±3.05	57.04±4.83	92.86±5.51	97.04±3.16	67.37±7.37	99.01±1.32
5	48.28±14.61	86.11±4.27	88.28±3.68	95.49±2.51	87.47±4.60	98.80±1.46
6	96.09±2.25	96.04±2.29	95.07±1.72	95.42±1.61	94.19±2.80	99.25±0.61
7	**0.00±0.00**	55.20±15.75	66.80±26.27	92.80±7.96	63.20±19.94	99.60±1.26
8	98.91±0.45	98.86±0.68	97.33±2.00	98.81±1.54	98.53±0.84	99.98±0.07
9	**0.00±0.00**	**11.67±7.61**	**7.78±9.15**	**65.00±18.15**	**18.33±16.15**	**92.78±8.71**
10	30.00±9.07	78.11±2.23	79.75±4.86	91.95±3.34	72.31±2.53	98.51±0.84
11	90.11±3.48	87.97±1.76	90.45±2.11	97.35±0.89	91.29±1.83	96.62±0.96
12	11.78±6.30	66.01±3.84	73.07±3.37	93.93±2.72	67.64±3.75	97.38±1.52
13	94.00±1.38	96.86±1.34	98.22±1.82	99.46±0.62	96.59±3.05	99.30±0.72
14	98.45±0.36	96.18±0.62	95.01±1.58	98.54±1.02	94.24±1.17	99.90±0.17
15	8.76±3.77	57.12±2.73	85.76±4.26	97.38±1.74	60.58±1.91	99.80±0.33
16	83.10±2.56	86.07±3.72	94.76±4.60	92.98±4.14	93.93±3.82	95.71±2.46
R-range	98.91±14.61	87.19±15.13	90.44±24.69	34.46±17.53	80.20±19.10	7.20±8.64
OA	62.32±1.64	81.88±0.42	88.24±0.92	95.89±0.46	79.64±0.67	98.25±0.32
AP	59.26±2.31	82.58±1.86	82.81±3.14	94.40±1.68	81.15±1.90	97.66±0.53
AR	46.06±1.67	72.26±1.64	81.28±1.69	92.45±1.42	73.31±1.69	97.89±0.64
AF_1	46.10±1.97	74.91±1.46	81.16±1.47	93.14±1.30	75.14±1.37	97.72±0.49
TIME(s)	0.04±0.01	5.81±0.24	33.98±0.71	39.03±0.71	43.06±1.78	42.09±0.61



(a) (b)

Fig. 4. Ground truth of (a) Indian Pines and (b) Pavia University.

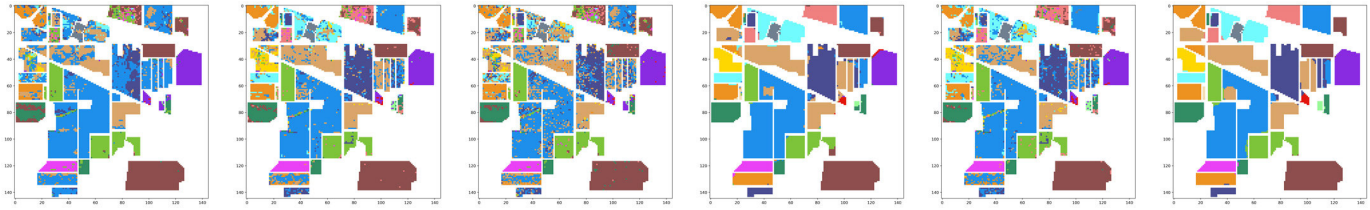
synthetic oversampling algorithm may synthesize samples with high repetition, and its effect will be reduced. And it is more likely to occur in the method that uses bootstrap for ensemble. However, by combining with the method of feature space optimization, the MinRecall of SSFRoS can still achieve 92.78%, while the R-range is 7.20%.

Table V and Fig. 6 show the experimental results of using different methods for the uncorrelated version of the Indian Pines dataset when the train size is 30% and 10%, respectively. The trends when the train size is taken as 20% are the same. A comparison of Table V with Table II and Fig. 6 with Fig. 5 indicates the noise robustness. It reveals that the method combined with 2-D-SSA for spatial feature extraction and fusion has better denoising ability, and the water absorption bands still contain usable information that is beneficial for classification.

Table VI and Fig. 7 show the experimental results of the Pavia University dataset when the train size is 10% and 1%, respectively. The trends when the train size is taken as 5% are the same, and the metrics of the proposed method at 1% were specifically shown in Table VII. The experimental results of these two datasets can corroborate each other, demonstrating that SSFRoS does improve the performance of imbalanced hyperspectral image classification.

It is worth noting that in the experimental results of the Indian pines dataset, the MinRecall of the methods except SSFRoS is mostly found in Class 9 Oats. Only 20 samples in the dataset belong to this class, which is the lowest. Class 7 Grass-pasture-mowed has only 28 samples, but it is not always the class with poor classification performance. In the experiments on the Pavia University dataset, the classes corresponding to the MinRecall of different methods are different. The sample numbers of frequently occurring classes, including Class 3 Gravel, Class 7 Bitumen, and Class 9 Shadows, are 2099, 1330, and 947, respectively. The sample size of Class 5 Painted metal sheets is only 1345, but its classification performance is even higher than the average. This proves that the imbalanced classification problem is not only caused by between-class imbalance, but is closely related to the data distribution within the classes. For example, classes with low overlapping degrees and long distances from other classes will be more easily classified, even if their training samples be relatively few, and will not lead to extremely low classification performance [11]. In addition to resampling, we may find an optimized space that is conducive to minority representation and the classification of different classes through feature extraction, feature selection, and feature rotation. Then, we can effectively solve or alleviate the imbalanced classification problem. Especially when the train size is small, although the synthetic oversampling step can add part of the available

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(a) (b) (c) (d) (e) (f)

Fig. 5. Classification maps by different methods on the corrected Indian Pines with train size is 10%. (a) SVM. (b) RoF. (c) 2-D-SSA. (d) SSARoF. (e) SSARoF. (f) SSFRoF.

TABLE V

EVALUATION METRICS (%) BY DIFFERENT METHODS ON THE UNCORRECTED INDIAN PINES DATASET WITH TRAIN SIZE = 30%

Class No.	SVM	RoF	2D-SSA	SSARoF	DSRoF	SSFRoS
1	**0.00±0.00**	60.63±11.52	96.25±4.37	95.00±3.95	62.19±6.82	97.81±2.57
2	53.62±2.57	79.64±0.88	97.72±1.09	98.75±0.79	68.62±1.70	99.76±0.28
3	42.29±2.01	69.04±1.42	98.52±0.92	99.12±0.60	64.78±1.70	99.86±0.20
4	13.37±3.93	71.39±5.65	99.22±1.10	99.88±0.25	74.64±5.11	99.76±0.51
5	79.32±2.85	90.98±1.93	97.72±0.78	98.88±0.62	91.36±1.68	100.00±0.00
6	97.77±0.65	97.59±0.65	98.96±0.46	98.32±0.56	96.95±0.88	99.73±0.25
7	**0.00±0.00**	82.50±8.90	95.00±7.07	97.50±3.54	81.00±10.49	100.00±0.00
8	99.25±0.35	99.40±0.42	99.94±0.19	100.00±0.00	99.43±0.55	100.00±0.00
9	**0.00±0.00**	**38.57±15.87**	**74.29±15.13**	**93.57±7.10**	**62.14±12.62**	100.00±0.00
10	57.54±4.48	82.54±1.23	97.65±0.96	99.32±0.56	76.54±1.10	99.90±0.10
11	89.71±1.58	90.80±1.05	98.84±0.61	99.49±0.41	92.84±0.85	97.89±0.61
12	32.82±3.65	76.02±2.09	95.20±1.99	98.72±1.17	76.99±1.36	99.16±0.66
13	94.48±1.49	96.71±2.72	99.51±0.58	99.72±0.49	95.87±2.39	99.79±0.47
14	97.95±0.39	96.96±0.61	99.62±0.26	99.79±0.15	95.62±1.11	99.95±0.06
15	33.63±1.41	65.04±3.82	99.37±0.55	99.22±0.69	68.56±2.68	99.74±0.70
16	87.23±1.78	93.85±2.99	95.38±3.08	94.46±4.42	96.77±1.69	**97.69±3.64**
R-range	99.25±4.48	60.83±15.45	25.65±14.94	6.43±7.10	37.29±12.07	2.31±3.64
OA	71.42±0.72	85.97±0.30	98.38±0.20	99.17±0.16	84.14±0.31	99.32±0.17
AP	64.75±0.66	89.00±1.17	96.48±1.08	98.90±0.43	87.96±1.21	99.23±0.35
AR	54.94±0.56	80.73±1.09	96.45±0.93	98.23±0.59	81.52±1.13	99.44±0.34
AF_1	56.37±0.56	83.55±1.08	96.33±0.68	98.53±0.43	83.91±1.03	99.32±0.27
TIME(s)	0.38±0.08	20.90±0.35	36.29±0.67	51.55±0.32	405.38±2.28	80.89±1.02

TABLE VI

EVALUATION METRICS (%) BY DIFFERENT METHODS ON THE PAVIA UNIVERSITY DATASET WITH TRAIN SIZE = 10%

Class No.	SVM	RoF	2D-SSA	SSARoF	DSRoF	SSFRoS
1	92.72±0.50	93.93±0.29	90.64±1.19	97.84±0.46	93.33±0.53	99.86±0.11
2	99.18±0.15	98.13±0.18	99.37±0.22	99.92±0.04	97.86±0.23	99.96±0.03
3	7.07±2.40	**75.75±1.50**	73.17±1.76	99.00±1.36	**76.49±1.73**	99.20±0.81
4	86.07±1.16	91.94±1.15	86.60±1.45	95.15±0.66	91.40±0.91	97.19±0.60
5	99.17±0.21	99.22±0.25	99.46±0.31	99.41±0.49	99.55±0.18	99.81±0.18
6	25.33±2.21	79.83±1.09	93.57±0.96	99.73±0.19	77.29±1.69	100.00±0.00
7	**0.00±0.00**	79.97±2.12	**66.42±3.16**	91.87±2.92	81.20±2.78	99.96±0.09
8	94.16±0.79	89.37±1.34	87.50±1.52	94.77±0.84	88.75±1.32	98.58±0.72
9	99.89±0.07	99.91±0.17	84.98±2.38	**91.01±1.70**	99.99±0.04	**95.81±1.50**
R-range	99.89±2.40	24.16±1.95	33.04±2.94	8.91±2.88	23.50±2.74	4.19±1.50
OA	80.54±0.22	92.54±0.17	92.77±0.25	97.84±0.20	92.03±0.25	99.50±0.09
AP	76.56±0.46	92.61±0.27	91.59±0.22	97.03±0.22	92.19±0.36	99.11±0.20
AR	67.07±0.27	89.78±0.35	86.86±0.61	95.52±0.45	89.54±0.43	98.93±0.17
AF_1	65.79±0.51	91.05±0.24	88.87±0.38	96.23±0.30	90.71±0.32	99.01±0.17
TIME(s)	0.21±0.02	11.61±0.29	188.24±3.66	198.13±3.48	914.61±17.48	251.73±3.74

information for minority classification, the effect is limited. The combination of sample space optimization and feature space optimization may achieve better results.

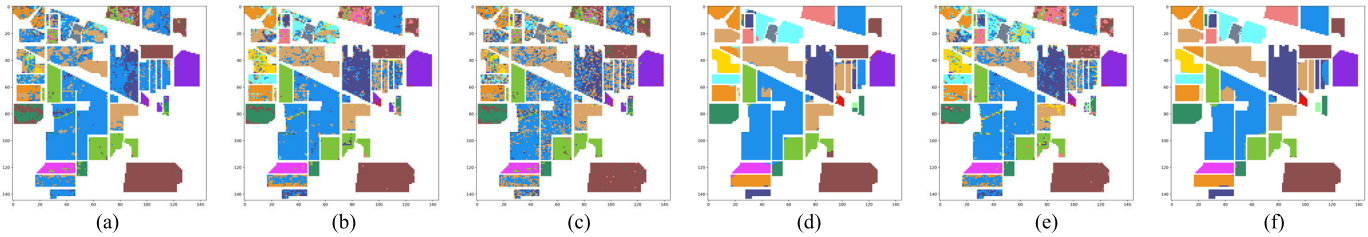
In addition, we also compared the newly proposed deep-learning method 3-D-HypeGAMO [31] for imbalanced hyperspectral image classification. Table VII shows the results from

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TABLE VII

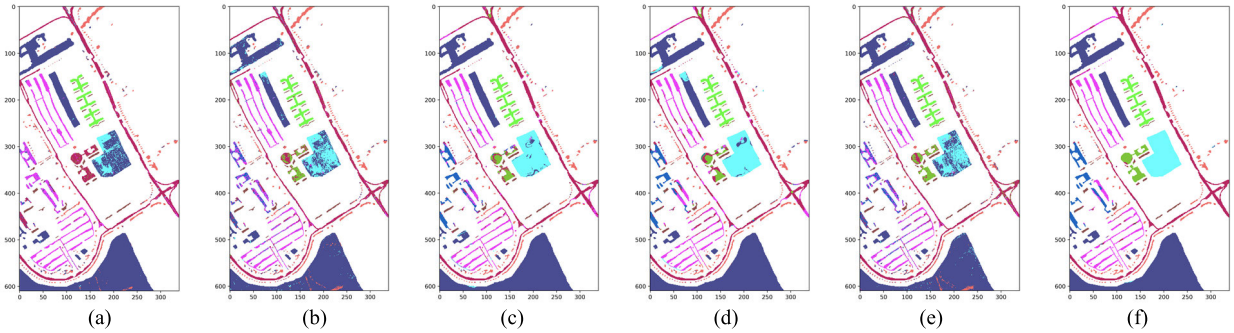
EVALUATION METRICS (%) BY THE DEEP-LEARNING METHOD AND PROPOSED METHOD ON THE CORRECTED INDIAN PINES DATASET WITH TRAIN SIZE = 10% AND THE PAVIA UNIVERSITY DATASET WITH TRAIN SIZE = 1%

Dataset	corrected Indian pines		Pavia University	
Class No.	HyperGAMO[32]	SSFRoS	HyperGAMO	SSFRoS
1	**93.33±9.73**	93.41±5.64	92.54±4.01	96.61±2.46
2	99.20±2.53	98.50±0.67	99.71±0.75	99.68±0.26
3	93.90±8.07	97.66±1.82	**72.50±11.41**	**86.54±7.04**
4	97.14±2.70	99.01±1.32	78.78±8.60	92.58±1.61
5	99.68±0.58	98.80±1.46	94.30±4.71	98.36±2.22
6	97.84±2.27	99.25±0.61	94.15±2.84	99.83±0.24
7	98.93±0.79	99.60±1.26	94.34±5.77	97.79±2.48
8	97.06±1.68	99.98±0.07	95.30±2.59	92.19±3.44
9	100.00±0.00	**92.78±8.71**	98.61±1.12	87.23±3.12
10	98.05±1.58	98.51±0.84		
11	99.39±0.57	96.62±0.96		
12	97.00±2.73	97.38±1.52		
13	96.58±2.50	99.30±0.72		
14	99.68±0.38	99.90±0.17		
15	98.61±1.23	99.80±0.33		
16	99.15±0.45	95.71±2.46		
R-range	6.67±9.73	7.20±8.64	27.21±10.66	13.29±6.80
OA	98.54±0.33	98.25±0.32	95.02±1.89	97.05±0.53
AP	98.05±0.72	97.66±0.53	94.77±2.00	95.99±0.58
AR	97.85±0.59	97.89±0.64	91.14±3.18	94.53±1.01
AF_1	97.86±0.53	97.72±0.49	92.57±2.64	95.15±0.71
TIME(s)	9776.70±252.48	42.09±0.61	2540.90±35.52	37.32±0.70



(a) (b) (c) (d) (e) (f)

Fig. 6. Classification maps by different methods on the uncorrected Indian Pines dataset with train size is 10%. (a) SVM. (b) RoF. (c) 2-D-SSA. (d) SSARoF. (e) DSRoF. (f) SSFRoF.



(a) (b) (c) (d) (e) (f)

Fig. 7. Classification maps by different methods on the Pavia University dataset with train size is 1%. (a) SVM. (b) RoF. (c) 2-D-SSA. (d) SSARoF. (e) DSRoF. (f) SSFRoF.

the experiments on the corrected Indian Pines dataset (train size = 10%) and the Pavia University dataset (train size = 1%) based on the data division and experimental settings in this article. It can be seen that the proposed method is competitive in all metrics. However, the proposed method has low demand for hardware and has great advantages in time complexity.

2) Ablation Experiments: To analyze the impact of each component, this section conducts ablation experiments on each part of the proposed method. The control methods are obtained by the following: (a) ablating the feature fusion component; (b) ablating the adaptive synthetic oversampling component; (c) replacing with the standard adaptive synthetic oversam-

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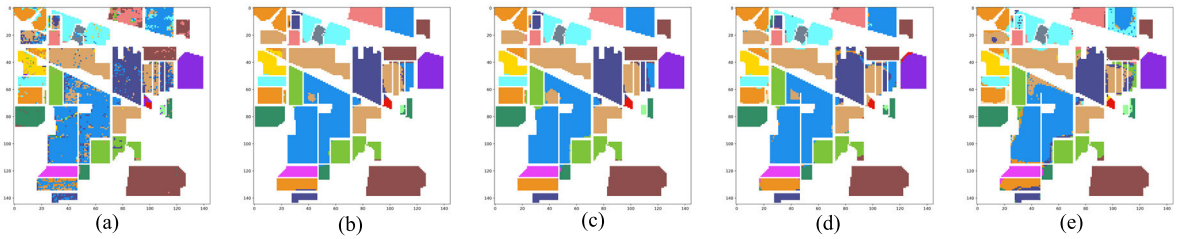


Fig. 8. Classification maps by different control methods on the corrected Indian Pines dataset with train size is 10%. (a) Ablating the feature fusion component. (b) Ablating the adaptive synthetic oversampling component. (c) Replacing with the standard adaptive synthetic oversampling component based on KNN. (d) Replacing with the standard RoF classifier. (e) Replacing with the PCA-based SVM ensemble classifier.

TABLE VIII
EVALUATION METRICS (%) BY DIFFERENT CONTROL METHODS ON THE CORRECTED INDIAN PINES DATASET WITH TRAIN SIZE = 10%

Class No.	a	b	c	d	e	SSFRoS
1	57.56±12.81	92.44±4.94	92.68±5.97	76.83±10.92	91.22±9.43	93.41±5.64
2	83.57±2.06	96.50±2.40	98.02±0.81	94.41±1.87	79.53±6.74	98.50±0.67
3	71.85±3.94	96.04±3.43	97.14±2.97	95.29±1.39	86.20±3.82	97.66±1.82
4	77.84±7.84	97.89±1.28	98.73±1.42	94.37±5.48	97.84±4.02	99.01±1.32
5	93.47±1.74	95.82±1.07	97.08±2.37	96.21±2.54	90.09±2.44	98.80±1.46
6	97.82±0.83	98.49±0.65	99.13±0.61	96.15±0.75	96.85±1.32	99.25±0.61
7	74.40±12.39	35.60±25.54	99.60±1.26	75.20±15.87	99.60±1.26	99.60±1.26
8	99.67±0.63	99.98±0.07	99.98±0.07	99.23±1.50	99.77±0.35	99.98±0.07
9	**39.44±23.78**	**0.00±0.00**	**46.11±48.86**	**50.56±16.45**	77.78±21.91	**92.78±8.71**
10	84.32±2.76	94.48±2.11	97.23±2.06	90.78±2.54	85.09±2.65	98.51±0.84
11	76.90±2.71	99.25±0.57	97.42±0.94	97.04±0.76	**76.28±3.40**	96.62±0.96
12	93.60±1.83	96.29±1.19	96.82±1.30	93.63±3.10	87.08±4.86	97.38±1.52
13	99.46±0.25	98.86±0.97	99.08±0.99	98.81±1.59	99.84±0.26	99.30±0.72
14	96.49±0.85	99.83±0.27	99.89±0.22	99.02±1.25	97.66±1.51	99.90±0.17
15	69.74±2.56	99.51±0.72	99.86±0.28	96.86±2.04	97.12±2.86	99.80±0.33
16	84.05±2.93	93.93±3.99	95.83±2.59	93.21±4.89	97.38±3.54	95.71±2.46
R-range	60.23±23.53	99.98±25.54	53.87±48.79	48.67±15.70	23.56±21.65	7.20±8.64
OA	84.95±0.57	97.40±0.51	97.98±0.49	95.68±0.61	86.93±1.37	98.25±0.32
AP	89.48±0.48	97.87±0.36	97.50±0.79	93.24±1.52	82.73±1.38	97.66±0.53
AR	81.26±1.88	87.18±1.36	94.66±3.24	90.47±1.39	91.21±1.61	97.89±0.64
AF_1	83.54±1.69	88.10±1.66	94.47±3.14	91.41±1.38	85.64±1.40	97.72±0.49

pling component based on KNN; (d) replacing with the standard RoF classifier; and (e) replacing with the PCA-based SVM ensemble classifier. Fig. 8 and Table VIII show the classification maps and metrics by different control methods in the corrected Indian Pines (train size is 10%), and the experimental results on the Pavia University dataset (train size = 1%) are listed in Fig. 9 and Table IX.

By comparing each control method in Figs. 8 and 9 with Figs. 5(f) and 7(f) obtained by the proposed method, it can be seen that each component of SSFRoS has a positive effect on the improvement of classification performance. Among the methods, B (ablating the adaptive synthetic oversampling component) and C (replacing with the standard adaptive synthetic oversampling component based on KNN) seem to have competitive performance on the classification maps. From the perspective of OA, the overall classification performance of these two methods is good, close to the performance of the proposed method. However, from the MinRecall and *R*-range, the method combined with the improved synthetic oversampling algorithm in this article can greatly improve the recognition performance of hard-to-learn classes and narrow the performance differences among different classes. Hard-to-learn classes refer to those that cannot be well learned by the classifier and have a large gap in recognition performance with other classes without oversampling or using standard KNN-based synthetic oversampling steps. For example, the class-recall of control methods B and C on Class 9 Oats on the corrected Indian Pines dataset is only 0.00±0.00% and 46.11±48.86%, respectively, and the *R*-range is 99.98±25.54% and 53.87± 48.79%. While the proposed method reaches 92.78±8.71% on Class 9 (increased by 46.67% compared to the standard algorithm) and reduces the *R*-range to 7.20±8.64%. In the Pavia University dataset, the recall of Class 3 Gravel increased by 5.26%. Figs. 10 and 11 compare method C and the method proposed in this article to show the improvement of classification performance of hard-to-learn classes. From the perspective of average metrics, the proposed method can improve recall and the F_1 score while maintaining precision. In addition, due to the random cross-validation of a hyperspectral image in the experiment in this article, the training set is not a complete image, and there is information loss in the utilization of the spatial neighborhood. When the complete image is used as the training set, better results may be achieved.

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TABLE IX
EVALUATION METRICS (%) BY DIFFERENT CONTROL METHODS ON THE PAVIA UNIVERSITY DATASET WITH TRAIN SIZE = 1%

Class No.	a	b	c	d	e	SSFRoS
1	85.94±3.71	95.64±2.39	96.12±2.44	90.80±3.60	**66.73±5.18**	96.61±2.46
2	89.43±3.71	100.00±0.01	99.84±0.17	99.01±0.80	78.81±4.04	99.68±0.26
3	**68.36±3.91**	**66.77±5.32**	**81.28±6.86**	77.12±2.89	73.09±7.21	**86.54±7.04**
4	90.98±3.74	85.00±3.40	91.36±1.73	88.18±3.01	91.48±1.90	92.58±1.61
5	99.61±0.13	97.67±2.53	98.09±2.45	98.27±3.20	96.20±7.27	98.36±2.22
6	84.76±3.10	95.44±2.17	99.34±1.20	93.18±1.65	95.96±1.58	99.83±0.24
7	79.73±4.59	87.05±5.52	92.98±5.18	**60.77±12.82**	78.33±4.96	97.79±2.48
8	83.19±4.30	88.08±3.04	91.37±2.79	80.67±5.32	73.46±6.38	92.19±3.44
9	99.55±0.34	71.05±19.17	84.76±4.52	75.93±3.11	81.13±3.68	87.23±3.12
R-range	31.25±4.46	33.23±19.16	18.56±6.69	38.24±12.02	29.47±5.69	13.29±6.80
OA	87.12±1.24	93.94±0.92	96.36±0.85	91.90±0.54	79.70±1.73	97.05±0.53
AP	85.04±1.46	94.35±1.10	95.61±0.87	90.28±1.20	76.87±2.08	95.99±0.58
AR	86.84±0.58	87.41±2.86	92.79±1.78	84.88±1.40	81.69±1.59	94.53±1.01
AF_1	85.66±1.08	89.97±2.63	93.99±1.38	86.98±1.29	77.85±1.89	95.15±0.71

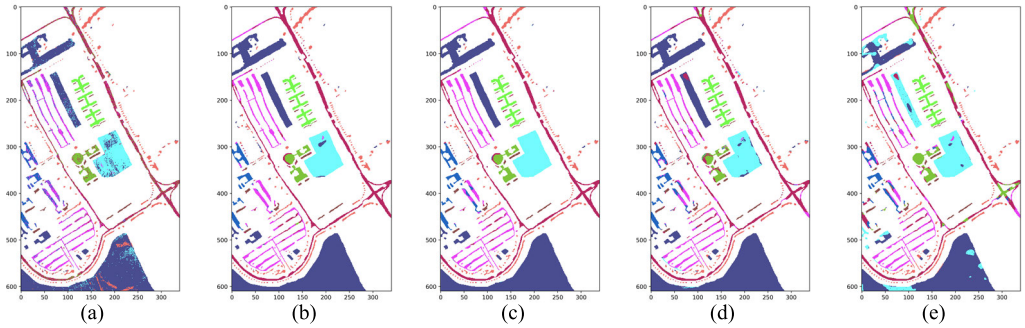


Fig. 9. Classification maps by different control methods on the Pavia University dataset with train size is 1%.

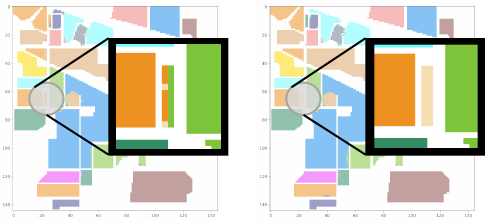


Fig. 10. Classification maps of hard-to-learn class Oats on the corrected Indian Pines dataset with train size is 10%. (a) Control method replacing with the standard adaptive synthetic oversampling component based on KNN. (b) SSFRoS.

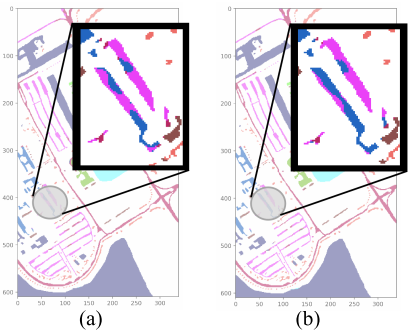


Fig. 11. Classification maps of hard-to-learn class Gravel on the Pavia University dataset with train size is 1%. (a) Control method replacing with the standard adaptive synthetic oversampling component based on KNN. (b) SSFRoS.

To measure the impact of each method on the class overlapping problem, this article uses the proposed distance metric Dis to quantify the severity of the class overlapping problem. We compare the rotary feature space solved by the method proposed in this article with method D (replacing it with the standard RoF classifier). The specific experimental settings are as follows: the same random tenfold cross-validation data are used, the number of base classifiers is set to ten, the ten rotation matrices are used to perform feature space rotation on the test data in an experiment, and then the average distance metric Dis is calculated in 100 spaces. The Dis of the control method D is 0.0767, which shows little difference as compared with the calculation results in the original feature space and is only reflected in the 16th decimal place or later. The method proposed in this article is 0.0917, which shows that the optimized feature space determined by the proposed method can effectively alleviate the problem of class overlapping.

3) Experimental Analysis of Parameter Settings: For improved classification methods based on RoF, the number of base classifiers B and the number of split subsets G are two important parameters. Grid search experiments with parameters reveal that the classification performance does not grow monotonically with increased complexity. Fig. 12 shows the change in mean average recall (mAR), mean average F_1 score (mAF_1), and mean average precision (mAP) of the tenfold stratified cross-validation experiments on the dataset of the corrected Indian Pines (train size is 10%).

It is clear from Fig. 12 that the number of subsets G has a greater impact on the classification performance. It shows that the classification performance may first improve and then

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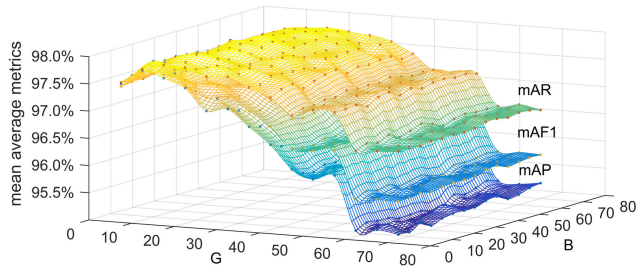


Fig. 12. Trend of mean average metrics in grid search experiments for parameters B and G on the corrected Indian Pines dataset (train size is 10%).

decrease as the number of subsets increases. This may be related to the fact that too small subsets would make it difficult to describe the sample with low-dimensional features. For both the uncorrected and corrected Indian Pines datasets, the method achieves the best performance at around $G = 15$. However, for the combinations with better performance, the value of B on the uncorrected dataset increases compared to that on the corrected dataset, which also reflects that the bagging-based ensemble method does reduce the overall error rate by reducing the variance. The trend of mean average metrics on the Pavia University dataset shows that the method achieves better performance at $G = 10$. From the analysis of the algorithmic properties of SSFRoS, this may be due to the fact that when the number of features f per subset is comparable to the number of classes c, using the projection space found by the proposed method can better preserve the information of the original data. In addition, increasing the number of B has a greater impact on the complexity. Therefore, in our experiments, we choose $B = 10$, when the ensemble size is small and classification performance is relatively good. When applying the proposed method to a dataset that may have large noise, appropriately increasing B at the cost of complexity can be considered in exchange for better classification performance.

For the imbalanced classification problem, the use of a resampling algorithm for balancing samples between classes requires setting two parameters, including the imbalance threshold and the sampling rate. Experiments demonstrate that the imbalance threshold has a greater impact on classification performance and running time. Most combinations of parameters with thresholds above 0.6 are competitive when experiments are conducted on the Indian Pines dataset (train size is 10%), while most combinations with thresholds below 0.4 are significantly worse when experiments are conducted on the Pavia University dataset (train size = 1%). The algorithm runtime grows with increasing imbalance threshold and sampling rate and has a roughly linear positive correlation with the imbalance threshold. It is also found that the algorithm performance is robust to some extent to the settings of the two parameters, and good performance is achieved when the imbalance threshold is not excessively low. As mentioned above, the between-class imbalance in the number of samples is not the only factor that determines the performance of imbalance classification. Using a fully balanced training set does not necessarily achieve better results than an imperfectly balanced training set. Besides, more synthetic samples need to be generated due to the increased threshold and sampling rate, which will increase the training time and the risk of data overfitting. Therefore, the imbalance threshold and sampling rate cannot be simply set to one. Considering the generalizability of the algorithm, we set the imbalance threshold and sampling rate as 0.8. They should be adjusted in practice according to the overlapping degree and the requirements for algorithm running time.

V. Conclusion

To deal with the imbalanced hyperspectral image classification problem, this article proposes an improved spectral–spatial feature rotation based on the SVM ensemble method (SSFRoF). It is a three-stage method including data preprocessing (feature fusion and resampling), feature rotation, and ensembled classification. In the data preprocessing stage and from the perspective of sample space optimization, this article proposes an SSAS algorithm. It not only considers the nearest neighbors based on KNN, but also counts the spatial neighbors in the training set. The samples surrounded by heterogeneous samples are more likely to be selected as seed samples, and the synthetic samples generated by interpolation can carry the information of adjacent samples in the space. In the feature rotation stage and from the perspective of feature space optimization, we innovatively discover that the class overlapping problem can be handled by combining feature extraction. First, we split the features into subsets randomly and then find the optimized feature subspace with a smaller within-class distance and larger between-class distance. The proposed rotation matrix solution method is used to obtain the final optimized rotation space to achieve the goal of reducing the difficulty of imbalanced classification. At the ensembled classification stage, SVM is used as the base classifier and the final classification results are obtained by voting.

The comparative and ablation experiments conducted on three versions of two public hyperspectral image datasets in this article show that.

1) The SSAS algorithm effectively uses the spatial information between hyperspectral pixels in two steps. It can further improve the overall classification performance and imbalanced classification performance, and has strong noise robustness. In the ablation experiment, the metrics indicate that the use of the improved spectral–spatial neighborhood based synthetic oversampling step can add useful information to the learning of most classes (especially on hard-to-learn classes which achieve low performance without oversampling or using standard KNN-based synthetic oversampling step) and effectively reduce the performance imbalance between-classes. The algorithm also has parameter robustness to some extent, and it is suggested that the imbalance threshold needs to be set properly according to the data complexity. However, feature fusion requires additional time in practical applications.
2) The feature space rotation and ensemble method proposed in this article can provide a diversified optimization feature space to alleviate the class overlapping

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problem. It further improves the performance metrics of imbalanced classification on the basis of standard RoF and has strong stability. The quantitative experiment of the severity of the class overlapping problem in the ablation experiment demonstrates that the optimized feature space after rotation can effectively alleviate class overlapping commonly found in the real distribution. Thus, the goal of reducing the difficulty of imbalanced classification is achieved. The number of subsets has a relatively greater impact on classification performance and better performance can be realized when the number of features in each subset is comparable to the number of classes.

Currently, there is no distinct mathematical definition and recognized quantitative standard for various complex distribution problems such as small disjuncts and class overlapping. In addition to using classical feature extraction algorithms to alleviate the class overlapping problem, it may be a sensible idea in the future to propose a new method to optimize the feature space calculation, or to interfere from the perspective of small disjuncts and other with-in-class complex distribution problems.

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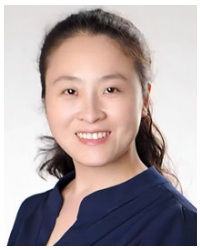
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